



27-11-1998
Dr. en Physique



French
Native



Serbian
Native

+



English
Fluent C1



German
Standard B2



Korean
Elementary A2

EDUCATION

BINATIONAL PHD IN PHYSICS

French and German

2020 – 2024 | UNIVERSITY OF STRASBOURG & KARLSRUHE INSTITUTE OF TECHNOLOGY (KIT)
UNISTRA PHD AWARD & SUMMA CUM LAUDE



MSC IN PHYSICS (120+21 ECTS)

Condensed Matter and Nanophysics
2018 – 2020 | UNIVERSITY OF STRASBOURG
Faculty of Physics and Engineering & IPCMS
Magistère in Fundamental Physics
QMat International Graduate School
Highest honours – Valedictorian

BACHELOR OF SCIENCE IN PHYSICS (180+12 ECTS)

2015 – 2018 | UNIVERSITY OF STRASBOURG
Faculty of Physics and Engineering
Magistère in Fundamental Physics
Honours

ASC IN MATHEMATICS

Lycée Kléber + MPA Bachelor
2015 – 2017 | UNIVERSITY OF STRASBOURG
Faculty of Mathematics and Computer Science
Honours

GENERAL SCIENTIFIC BACCALAUREATE

2013 – 2015 | LYCÉE J. MERMOZ, SAINT-LOUIS
Highest honours

CONTACT

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Seoul, KR

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10 cité du stade, 68300
Saint-Louis, FR

Denis Janković

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POSTDOCTORAL RESEARCHER

- PhD in Physics from the University of Strasbourg.
- Dr. rer. nat. from the Faculty of Chemistry and Biosciences at Karlsruhe Institute of Technology.
- Master's degree in Condensed Matter and Nanophysics.
- Lindau Alumni | 73rd Lindau Nobel Laureate Meeting in Physics.

EXPERIENCE

2025-present Center for Quantum Nanoscience, Institute for Basic Science & Ewha Womans University (Seoul, KOR)
Postdoctoral Researcher



Theory of Quantum Spins on Surfaces Group

Study of electron- and nuclear-spin qubits on surfaces using quantum control and open-system simulations, supporting ESR-STM experiments.

Keywords: Quantum information, Optimal control, PINN, Open systems, ESR-STM

2020-2024



IPCMS (Strasbourg, FR) & INT-KIT (Karlsruhe, DE)
International co-tutelle PhD

Supervisors: Pr. Dr. Paul-Antoine Hervieux (IPCMS) & Pr. Dr. Mario Ruben (KIT)

Hyperfine interactions in lanthanide-organic complexes for quantum information processing.

Calculate and diagonalize the valence-electron Hamiltonian of an Ln ion, whose nucleus is accessible through the hyperfine interaction. Obtain a model of parameters useful for optimizing the optical or electrical manipulation of nuclear-spin qubits, e.g. the ligand (geometry, composition), pulse shapes, etc.

Keywords: Quantum information, Optimal control, Open systems, Hyperfine interaction

2020



IPCMS (Strasbourg, FR)

Department of Ultrafast Optics and Nanophotonics
Master's internship

Supervisor: Pr. Dr. Paul-Antoine Hervieux

Keywords: Ligand fields, Atomic physics, Racah algebra, Hyperfine interaction.

2019



Faculty of Physics - University of Basel (CH)

Condensed Matter Theory & Quantum Computing Group
Master's internship

Supervisors: Pr. Dr. Daniel Loss - Pr. Dr. Marko Rančić

Keywords: Condensed matter, Numerical simulation, Quantum computing

MAIN PUBLICATIONS

06/26



Probing the Weak-Driving Quantum Speed Limit via Drift-Aware Shooting Methods

Preprint: [arXiv:2606.22212](https://arxiv.org/abs/2606.22212)

Janković, D., Taherpour, S., Etienney, PL., Hervieux, PA. and Wolf, C.

03/26



Overcoming limitations on gate fidelity in noisy static exchange-coupled surface qubits

DOI: 10.1038/s41534-026-01214-1

Le, HA., Taherpour, S., Janković, D. and Wolf, C.
npj Quantum Inf 12, 69 (2026).

04/25



Nonlinearity of the fidelity of open qudit systems: Gate and noise dependence in high-dimensional quantum computing

DOI: 10.22331/q-2025-04-07-1690

Hartmann, JG., Janković, D., Pasquier R., Ruben, M. and Hervieux, PA. *Quantum* 9, 1690 (2025).

06/24



Noisy qudit vs multiple qubits : Conditions on gate efficiency for enhancing fidelity

DOI: 10.1038/s41534-024-00829-6

Janković, D., Hartmann, JG., Ruben, M. and Hervieux, PA. *npj Quantum Inf* 10, 59 (2024).



@G Scholar

FELLOWSHIPS



**2021 - Franco-German University co-
2024 tutelle support grant**

**2018- "Q-Master fellowship" - QMat
2020 Interdisciplinary Thematic
Institute**

**2015- CROUS merit- and needs-
2018 based scholarship**

TRAINING

Technical

Parallel Computing

21 h - University of Strasbourg - 2023

Group Theory Course

21 h - GROUP34, Strasbourg - 2022

Quantum Information Spring School

43 h - École de Physique des Houches -
2022

39th Joliot-Curie International School

25 h - Île d'Oléron - 2021

Transferable Skills

English for Scientific Writing

20 h - University of Strasbourg - 2023

Scientific Writing Methodology

7 h - University of Strasbourg - 2023

Public Speaking

8 h - University of Strasbourg - 2022

Time Management

8 h - University of Strasbourg - 2021

Science Communication and Outreach

21 h - University of Paris-Saclay - 2020

Research Integrity

13 h - University of Strasbourg - 2020

**+ >100 h of multidisciplinary
training**



Denis Janković

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MAIN TEACHING

**2022- Tutorials in Physics for Life Sciences
2023 Teaching assistant (including grading and invigilation)**
16 hours at undergraduate level



**2021- Physics problem classes
2024 Adjunct instructor**
80 hours at undergraduate level



**2021- Experimental Physics Laboratory Classes
2024 Teaching assistant (including grading)**
97 hours at undergraduate level



**2020- Coordination of the online transition of the Magistère in
2021 Physics (COVID-19) (lectures, exam bank, tutorials)**
64 hours



**2015- Private tutoring for middle- and high-school students
2019 Physics, Mathematics, Life and Earth Sciences, Methodology**



DISTINCTIONS

06/25 2024 PhD Award from the University of Strasbourg
Society of Friends of the Universities of the Strasbourg Academy

09/23 European Summer School in QST 2023 (KIT, Durbach, DE)
Best poster award: 1st place

**09/21 Paris International School on Advanced Computational
Materials Science 2021 (Sorbonne Université, Paris, FR)**
Best poster award: 2nd place

09/20 KSOP-QMat Summer School 2020 (KIT, Karlsruhe, DE)
Prix du meilleur poster Matériaux & Dispositifs Photoniques

SKILLS

Scientific Programming

- Python, C, C++, Fortran, Matlab, Mathematica
- UNIX, HPC, OpenMP, MPI, CUDA, PyTorch, QuTiP

Scientific Service

- Reviewer for IOP Publishing (4 reviews)
- Doctoral selection committee for QuanG2 / QuantAlps / QuanTEdu-France (2026)

Community Involvement

- President of the Young Investigators Group (2018-2021)
- Student representative of the EUR QMat
- International Doctoral Programme - Delegate for the Primo Levi 2021 cohort

Scientific Events

- Student organizer of the online conference Machine Learning for Quantum X 2021
- Student organizer of the European Spring School in Quantum Science and Technology 2022

Layout and Editing

- LaTeX, Office (Word, Excel, PowerPoint, Visual Basic), Canva
- Blender, Adobe CC (Photoshop, After Effects, Premiere)

HOBBIES & INTERESTS

Scientific

Quantum information and optics, Atomic and molecular physics, Optimization & ML, Group theory, ...

Miscellaneous

Science communication, Video editing, Bouldering, Kendo/Kumdo, Cycling, Linguistics, Vexillology, Korean studies, Geology